Question #3: Flowchart

Objective: Selection, Loop, Expression from flowchart

Create a Python program to follow the program flow provide in given flowchart on next page.

INPUT

Many lines.

OUTPUT

One line as shown in examples

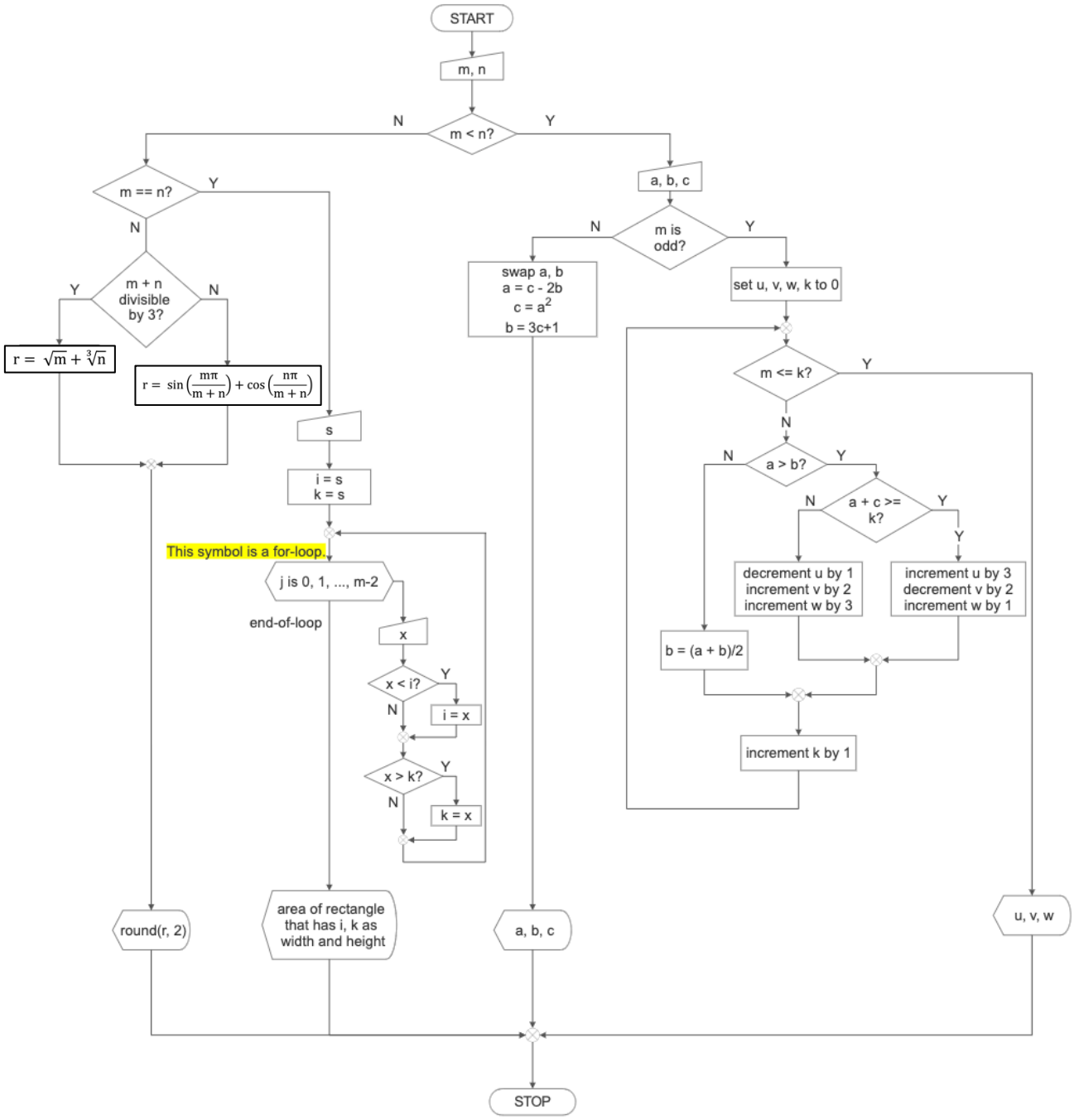
EXAMPLES

Input (from keyboard)	Output (on screen)	Explanation
3 6 6 4 2	9 -6 3	Each example shows input and expected output for all flow path of the flowchart.
4 7 2 4 6	2 13 4	
5 5 5 3 8 6 2	16	
7 4	1.33	
7 5	4.36	

TESTCASES in Grader

Testcases will be grouped. Each group has the following criteria:

Testcases quantity	Test case characteristics
20%	Flowchart path 1
20%	Flowchart path 2
20%	Flowchart path 3
20%	Flowchart path 4
20%	Flowchart path 5


START
m, n
m < n?
N
Y
m == n?
Y
N
m + n divisible by 3?
Y
N
$r = \sqrt{m} + \sqrt[3]{n}$
$r = \sin\left(\frac{m\pi}{m+n}\right) + \cos\left(\frac{n\pi}{m+n}\right)$
s
i = s
k = s
This symbol is a for-loop.
j is 0, 1, ..., m-2
end-of-loop
x
x < i?
i = x
x > k?
k = x
round(r, 2)
area of rectangle that has i, k as width and height
a, b, c
m is odd?
swap a, b
a = c - 2b
$c = a^2$
b = 3c+1
a, b, c
set u, v, w, k to 0
m <= k?
a > b?
a + c >= k?
decrement u by 1
increment v by 2
increment w by 3
increment u by 3
decrement v by 2
increment w by 1
b = (a + b)/2
increment k by 1
u, v, w
STOP